Algorithm 1:
Genetic optimization algorithm.

Step 1: Start

Step 2: Initialise the population, considering the chromosome as the population for this algorithm.

Step 3: Fitness evaluation - choose the best possible fit in the population of chromosomes using the fitness function.

Step 4: Selection - selecting the best suitable chromosomes to be the parents that would generate the new population or new generation.

Step 5: Crossover - crossing over the two parents to reproduce the new ones.

Step 6: Mutation - make small possible changes in the crossover to obtain more optimal solution.

Step 7: Termination - Repeat the process until best optimal solution is obtained that is the offspring

Step 8: Use the result to produce the offspring.

Machine learning Feature Selection.

Select useful i/p features for classification model.
Improve prod. performance by reducing number of features
In machine learning using large number of input features can increase model complexity, computational cost and training time.
The objective is to identify most relevant input features for a classification model, this improves classification accuracy and overall model performance.

→ Pseudocode.

Start.

Load the dataset.

Initialize a random population of features of input where 1 represents selected feature ~~feature~~ and 0 represents unselected feature.

Evaluate the fitness of each chromosome ~~and~~ (number of selected features)

WHILE termination condition is not satisfied DO.

SELECT the best suitable parent chromosome (that is previous best feature) for classification model.

Apply mutation to introduce variation in feature selection.

Calculate the fitness of the feature.

Form a new population (feature)

END WHILE.

Select the chromosome with highest fitness value.

Identify optimal subset of input features.

Train the model using selected features.

Evaluate and display selected features.

STOP.

Write c o/p seen.

→ Output.

Original number of features : 20.

Original features considered : ['feature_0', 'feature_1', 'feature_2', 'feature_3', 'feature_5', 'feature_6', 'feature_7', 'feature_8', 'feature_9', ...].

Reduced number of feature : 20.